

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457183.7451 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.042
Transit depth (Rp/Rs)^2	2.03 +/- 1.19 [%]
Orbital Inclination (inc)	83.06 +/- 2.79
Airmass coefficient 1 (a1)	0.47 +/- 0.39
Airmass coefficient 2 (a2)	0.42 +/- 0.2
Scatter in the residuals of the lightcurve fit is	69.02 %
Best Comparison Star	None
Optimal Aperture	3.01
Optimal Annulus	3.51
Transit Duration (day)	0.11 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.042 vs 0.10538 (deviation 0.47σ)
Duration fit vs published	0.11 d vs 0.1167 d (deviation 0.15σ)
Mid-time O–C	4.7 min
Depth SNR	1.8
Residual scatter	69.02 %

Light curve

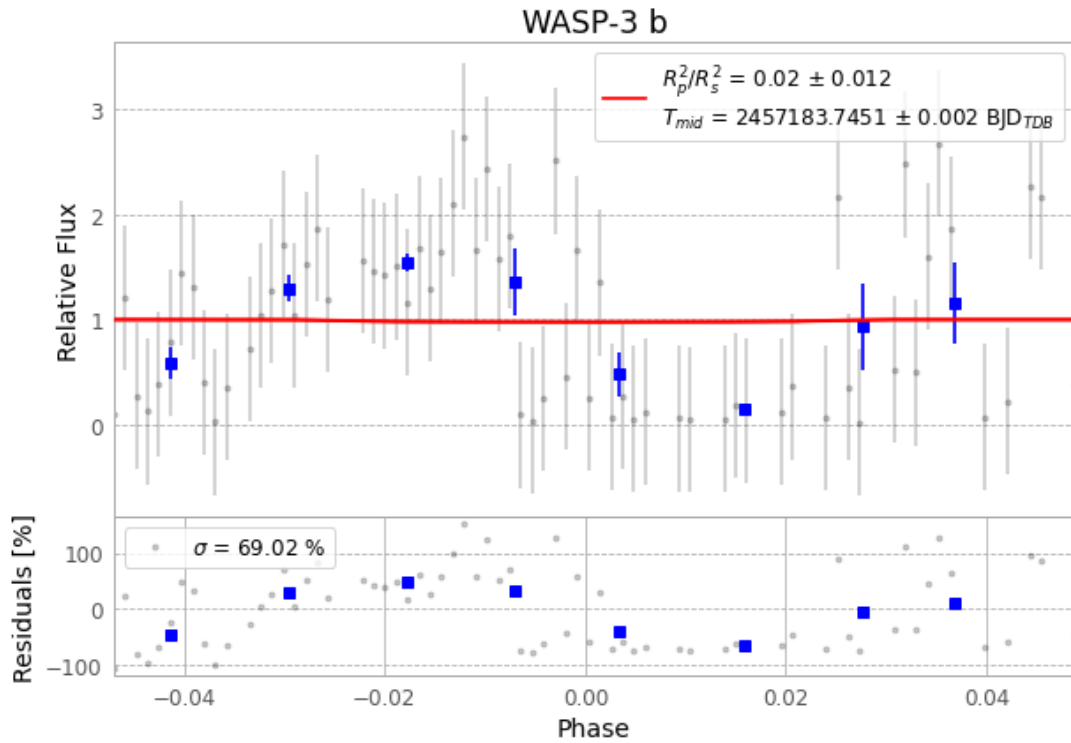


Figure 1 — FinalLightCurve_WASP-3 b_2015-06-09.png (SHA-256 50132e2ad9d77b9b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 310], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b91fd7152ee07078...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), WASP-3150610033912.FITS → WASP-3150610075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150610.log (877c5a9a6f45...), FinalLightCurve_WASP-3 b_2015-06-09.png (50132e2ad9d7...), FinalLightCurve_WASP-3 b_2015-06-09.csv (5dccd21cdf1e...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:15Z.